Real-Time ESG Scoring Using Satellite Imagery and Alternative Data

A Thesis submitted by

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DECLARATION

I declare that the project **"Real-Time ESG Scoring Using Satellite Imagery and Alternative Data"** is my work.

1. I confirm that the work in this document is based on my study and my code.
2. I confirm that the design and results appear first in this report.
3. I state that this work was not submitted to any institute or university for any degree or award.
4. I submit this work under the supervision of Dr. Puneet Sharma.

Signature

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CERTIFICATE

This is to certify that the Project Report *"Real-Time ESG Scoring Using Satellite Imagery and Alternative Data"*, submitted by Jojo Joseph (M24DE3041) to the Indian Institute of Technology Jodhpur for the award of the degree of Master of Technology in Data Engineering, is a bonafide record of the work carried out by him under my supervision. To the best of my knowledge, the contents of this report have not been submitted to any institute or university for any degree or diploma.

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Dr. Puneet Sharma

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ABSTRACT

ESG criteria have emerged as crucial benchmarks for evaluating how companies perform on sustainability and ethical fronts. Yet current assessment methods have significant limitations. Most ratings still depend heavily on what companies choose to disclose in their annual reports and periodic audits. This creates obvious problems: information arrives late, self-reporting introduces bias, and there's little real-time visibility into what's actually happening at corporate facilities.

This Phase 1 report tackles these issues by designing a real-time ESG scoring system that combines satellite imagery with alternative data sources and modern deep learning approaches. Rather than waiting for quarterly disclosures, the system can continuously monitor environmental indicators visible from space while incorporating news sentiment, corporate registry updates, and governance events as they happen.

The technical approach brings together several data streams that haven't been integrated before. Satellite imagery from sources like Sentinel-2 provides direct observation of corporate facilities—we can see emissions patterns, land use changes, and industrial expansion without relying on self-reports. To make sense of these images, we employ Vision Transformers, which have recently shown remarkable performance on remote sensing tasks. These visual features get combined with structured company data through Graph Neural Networks (specifically GraphSAGE and GAT architectures) that model how ESG risks propagate through sectoral relationships and supply chains.

Our literature review covers existing ESG methodologies and their documented shortcomings, remote sensing applications in sustainability monitoring, recent advances in vision transformers and graph neural networks, and multi-modal learning frameworks. We formulate ESG scoring as a data fusion challenge with several technical hurdles: limited ground-truth labels, temporal dynamics in corporate behavior, and the computational demands of processing satellite imagery at scale.

The proposed architecture uses event streaming (similar to Apache Kafka) to update scores incrementally as new data arrives. Rather than a black-box rating, the scoring engine produces transparent subscores for Environmental, Social, and Governance dimensions—each on a 0–100 scale with clear attribution to underlying features.

During Phase 1, we evaluated available data sources including free satellite platforms (Sentinel-2, Landsat-8/9), alternative data APIs (OpenCorporates, news aggregators), and deep learning frameworks (PyTorch ecosystem, Hugging Face models). Initial prototyping confirmed the approach is technically feasible while revealing specific challenges around data acquisition, feature fusion strategies, and interpretability requirements.

Phase 2 will implement the complete pipeline: data ingestion, ViT-based feature extraction from satellite images, GNN embedding generation, and real-time scoring. We'll validate first on synthetic data to establish baseline performance, then integrate real satellite imagery and test on actual companies. Success metrics include scoring accuracy, update latency, interpretability for stakeholders, and computational efficiency.

This work contributes to computational sustainability by showing how deep learning and real-time data streams can make ESG assessment more transparent and objective—ultimately supporting better investment decisions and corporate accountability.

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List of Abbreviations

API	Application Programming Interface
ATS	Applicant Tracking System
ANOVA	Analysis of Variance
CI	Confidence Interval
CNN	Convolutional Neural Network
CLS	Classification Token
CSV	Comma-Separated Values
E	Environmental (ESG dimension)
ESG	Environmental, Social, and Governance
G	Governance (ESG dimension)
GAT	Graph Attention Network
GNN	Graph Neural Network
GPU	Graphics Processing Unit
JSON	JavaScript Object Notation
MSCI	Morgan Stanley Capital International
NLP	Natural Language Processing
RGB	Red, Green, Blue
ReLU	Rectified Linear Unit
S	Social (ESG dimension)
SAE	Simultaneous Authentication of Equals
SHAP	SHapley Additive exPlanations
ViT	Vision Transformer

Real-Time ESG Scoring Using Satellite Imagery and Alternative Data

1 Introduction and Background

1.1 Motivation

ESG criteria have become central to how investors and regulators evaluate corporate performance. It's no longer just about quarterly earnings—investors managing trillions in assets now look at whether companies are sustainable, socially responsible, and well-governed [1]. The shift makes sense: companies that perform well on ESG dimensions tend to create better long-term value and manage risks more effectively. Governments worldwide have responded by mandating ESG disclosures, driven by mounting concerns about climate change, inequality, and corporate governance failures [2].

But here's the problem: how we currently measure ESG performance has serious flaws. Major rating agencies like MSCI, Sustainalytics, and Refinitiv depend almost entirely on what companies report about themselves—annual sustainability reports, voluntary disclosures, and occasional third-party audits. This creates several issues:

- **Information Lag:** By the time ESG scores update (typically quarterly or annually), the information is already stale. If something changes at a facility today, it won't show up in ratings for months.
- **Disclosure Bias:** Companies naturally present themselves in the best light. Greenwashing is real, and there's no way to verify many claims without independent monitoring.
- **Limited Verification:** Third-party audits happen infrequently and cost a fortune. They can't possibly capture everything happening across a company's operations in real time.
- **Rating Divergence:** Perhaps most troubling, different rating agencies often disagree significantly about the same company. Berg et al. [3] found surprisingly low correlations between major providers—a sign that current methods lack robustness.

1.2 Alternative Data and Remote Sensing

What if we could verify ESG performance independently, using data that companies can't manipulate? Recent technology advances make this increasingly possible. Satellite imagery offers a direct view of what's happening at corporate facilities. Sentinel-2, Landsat-8/9, and commercial providers now capture high-resolution images frequently enough to track:

- Facility emissions and air quality through spectral analysis
- Land use changes and deforestation near operations

- Water pollution events in manufacturing zones
- Construction and expansion activities that signal growth or environmental impact

The key advantage? Satellites don't wait for annual reports. They observe continuously.

Beyond imagery, alternative data sources are opening new windows into corporate behavior. News sentiment, social media discussions, corporate registry updates, and supply chain networks reveal information about social and governance performance—often months before it appears in official disclosures [4]. These signals complement satellite observations, creating a more complete picture.

1.3 Deep Learning for Multi-Modal ESG Assessment

This project brings together satellite imagery and alternative data using modern deep learning to build a real-time ESG scoring system. The challenge is combining fundamentally different data types—images, text, structured company data—into meaningful scores. Here's our technical approach:

- **Vision Transformers (ViT):** These models [5], pre-trained on millions of images, can extract meaningful features from satellite imagery. They identify patterns in environmental indicators around corporate facilities without requiring us to manually specify what to look for.
- **Graph Neural Networks (GNN):** ESG risks don't exist in isolation—they propagate through industry sectors and supply chains. We use GraphSAGE [6] and Graph Attention Networks [7] to model these company relationships, learning how ESG information flows between connected firms.
- **Real-Time Streaming:** Rather than batch-processing data weekly or monthly, we employ event streaming (similar to Apache Kafka) to update scores incrementally as new information arrives. This enables truly continuous monitoring.

1.4 Research Objectives

The primary objective of this work is to design, implement, and evaluate a prototype real-time ESG scoring system that:

1. Integrates satellite imagery with alternative data sources in a unified framework
2. Employs Vision Transformers for environmental feature extraction from remote sensing data
3. Leverages Graph Neural Networks to model company relationships and enhance ESG embeddings
4. Implements real-time streaming to enable continuous score updates
5. Produces transparent, interpretable ESG scores with dimension-specific subscores (E, S, G)
6. Provides a modular architecture extensible to real-world data sources and production deployment

1.5 Report Organization

The remainder of this Phase 1 report is organized as follows: Section 2 reviews related work in ESG scoring, satellite remote sensing for sustainability, and deep learning for multi-modal data fusion. Section 3 formally defines the problem and states research objectives. Section 4 details the proposed pipeline architecture, including data ingestion, ViT feature extraction, GNN-based embedding generation, and real-time scoring. Section 5 describes initial explorations and preliminary investigations conducted in Phase 1. Section 6 outlines the work plan and timeline for Phase 2 implementation. Section 7 summarizes Phase 1 contributions and discusses next steps.

2 Literature Survey

2.1 ESG Scoring and Rating Agencies

Does ESG performance actually matter for business outcomes? Friede et al. [1] synthesized over 2,000 studies and found that yes, it does—ESG factors correlate positively with financial performance in most cases. Companies that score well on sustainability tend to perform better over the long term. Eccles and Serafeim [2] showed this effect is especially pronounced in stakeholder-intensive industries, where ESG leaders enjoy lower cost of capital and better stock performance.

So the case for measuring ESG is solid. But how reliable are current measurements? Not very, it turns out. Berg et al. [3] compared ratings from major providers—MSCI, Sustainalytics, Refinitiv—and found surprisingly low agreement. The same company might score well on one platform and poorly on another, depending on methodology choices, scope decisions, and weighting schemes.

This divergence problem points to a deeper issue: reliance on self-reported data. Companies control what they disclose and when. Moving toward objective, externally verifiable data sources seems essential for improving rating reliability.

2.2 Satellite Remote Sensing for Environmental Monitoring

Satellites are increasingly being used to monitor corporate environmental performance directly. Unger et al. [8] showed that Sentinel-2 imagery can detect methane emissions from oil and gas facilities at scale—something previously requiring expensive on-site equipment. Li et al. [4] took a different angle, using nighttime lights and vegetation indices visible from space to track economic activity and environmental degradation in manufacturing zones.

The infrastructure for this kind of analysis has matured rapidly. Google Earth Engine [9] now provides access to petabytes of satellite data with cloud-based processing power, democratizing geospatial analysis. Researchers have used it to track deforestation [10], monitor water quality [11], and map urban heat islands [12]. All of these environmental indicators connect directly to corporate ESG performance, yet most companies never report on them.

2.3 Vision Transformers for Image Analysis

Vision Transformers (ViT) shook up computer vision when Dosovitskiy et al. [5] showed that attention-based architectures could match or beat traditional convolutional neural networks—without using convolutions at all. The key innovation is self-attention: the model learns to focus on relevant parts of an image globally, rather than processing local patches in isolation.

This turns out to be especially useful for satellite imagery. Environmental patterns often involve relationships across different scales—a factory's emissions plume might extend kilometers, while specific facility features matter at meter resolution. ViT's global attention mechanism handles this naturally. Researchers have already fine-tuned pre-trained ViT models for land cover classification [13], crop map-

ping [14], and disaster assessment [15].

The practical advantage for our work is transfer learning. Large-scale pre-trained models from Hugging Face and Google Research have already learned general visual features from millions of images. We can adapt them to ESG-specific patterns without needing massive labeled datasets—a crucial benefit given the scarcity of ground-truth ESG labels.

2.4 Graph Neural Networks for Relational Learning

Companies don't exist in isolation—they're connected through supply chains, sectoral relationships, and ownership structures. Graph Neural Networks provide a way to model these connections and learn from them. Hamilton et al. [6] developed GraphSAGE, which creates node embeddings by aggregating information from a company's neighbors in the graph. Veličković et al. [7] extended this with attention mechanisms in GAT, allowing the model to weight different neighbors differently during information aggregation.

Financial applications have proven GNNs' value. They've been used for credit risk [16], fraud detection [17], and stock prediction [18]. For ESG assessment, the appeal is clear: risks and best practices spread through company networks. If a supplier faces environmental violations, that affects everyone downstream. If an industry peer adopts new governance standards, others may follow. GNNs can capture these propagation dynamics in a way that treating companies independently cannot.

2.5 Real-Time Streaming and ESG Monitoring

Modern data processing frameworks have made real-time analytics feasible at scale. Apache Kafka [19], Flink [20], and Spark Streaming [21] can handle continuous high-velocity data streams—exactly what's needed for real-time ESG monitoring. These technologies already power financial market surveillance [22], social media analysis [23], and IoT sensor networks [24].

Chen et al. [25] took a step toward real-time ESG by incorporating news sentiment and social media signals, showing that timely alternative data improves score responsiveness compared to quarterly updates. But their approach had gaps: no satellite imagery, no modeling of company relationship networks. Our work addresses both limitations by combining visual Earth observation with graph-based company modeling in a unified streaming architecture.

2.6 Multi-Modal Deep Learning

Combining different data types—images, text, structured data—into unified models has proven powerful across many domains. Baltrusaitis et al. [26] surveyed the landscape of multi-modal learning, covering fusion architectures and joint embedding techniques that let models learn from complementary information sources.

Sustainability applications show what's possible. Jean et al. [27] combined satellite imagery with household surveys for poverty mapping, achieving better predictions than either data source alone. Kus-

sul et al. [28] fused optical and radar remote sensing for food security assessment, leveraging the complementary information in different wavelengths.

Our work extends this multi-modal approach to corporate ESG by bringing together three information sources that haven't been integrated before: ViT-processed satellite features (environmental), GNN-encoded company networks (systemic risk), and structured alternative data (social and governance signals).

Study	**Data Sources**	**Methods**	**Limitations**
Traditional ESG Ratings	Self-reported disclosures	Rule-based scoring	Delayed, subjective
Unger et al. [8]	Sentinel-2 imagery	CNN models	Single modality
Chen et al. [25]	News & social media	NLP sentiment	No imagery or graphs
This Work	Satellite + alternative data	ViT + GNN + streaming	Proposed system

Table 2.1: Comparison of ESG assessment approaches in literature.

3 Problem Definition and Objectives

3.1 Problem Statement

Given a universe of companies $\mathcal{C} = \{c_1, c_2, \ldots, c_N\}$, the objective is to compute real-time ESG scores $s_i \in [0, 100]$ for each company c_i by integrating:

- **Satellite imagery** I_i: RGB images of corporate facilities from optical remote sensing platforms
- **Tabular features** $\mathbf{x}_i \in \mathbb{R}^d$: Structured attributes including sector, emissions intensity, board independence, controversy counts, etc.
- **Alternative data streams**: News sentiment, corporate registry information, governance events
- **Company relationship graph** $\mathcal{G} = (\mathcal{C}, \mathcal{E})$: A graph where nodes represent companies and edges $(c_i, c_j) \in \mathcal{E}$ connect firms with sectoral or supply chain relationships

The ESG scoring function should:

1. Produce interpretable subscores for Environmental (E), Social (S), and Governance (G) dimensions
2. Update scores in real-time as new data arrives via streaming events
3. Be transparent and auditable, with clear feature attributions
4. Generalize to unseen companies in the same industrial sectors

3.2 Technical Challenges

Several technical challenges arise in constructing such a system:

- **Multi-modal fusion:** Satellite images and tabular features exist in different representational spaces. Effective fusion requires learning joint embeddings that capture complementary information.
- **Graph construction:** Defining meaningful edges in the company graph is non-trivial. Edges based on sector similarity may be too coarse, while supply chain networks require proprietary data.
- **Temporal dynamics:** ESG factors evolve over time. The system must distinguish transient events (e.g., temporary emissions spikes) from persistent patterns.
- **Limited labeled data:** Existing ESG ratings are sparse, inconsistent, and potentially biased. Training supervised models requires careful consideration of label quality.
- **Computational scalability:** Processing high-resolution satellite imagery and large-scale graphs in real-time demands efficient architectures and hardware acceleration.

3.3 Research Objectives

This work addresses the above challenges through the following specific objectives:

1. **Design a modular multi-modal pipeline** that separates concerns: data ingestion, feature extraction, graph embedding, scoring, and visualization.
2. **Implement Vision Transformer-based satellite feature extraction** using pre-trained models from Hugging Face, leveraging transfer learning to encode environmental patterns.
3. **Construct company relationship graphs** based on sector affiliations and geographic proximity as proxies for ESG risk transmission.
4. **Train Graph Neural Network models** (GraphSAGE, GAT) to generate ESG-aware company embeddings by propagating information through the relationship graph.
5. **Develop a transparent scoring engine** that combines learned embeddings with tabular features through interpretable formulas, producing E/S/G subscores and overall ESG ratings.
6. **Implement real-time streaming simulation** using Kafka-like event queues to demonstrate continuous score updates in response to new satellite observations, news, and governance events.
7. **Validate the pipeline** initially on synthetic data and establish a foundation for future work with real data sources.

3.4 Scope and Limitations

This Phase 1 report focuses on architectural design and methodology formulation. The proposed approach will be implemented and validated in Phase 2. Specific anticipated limitations include:

- Initial prototype will operate on *synthetic data* generated to mimic real ESG-relevant information distributions
- No access to proprietary supply chain data; graph edges will be constructed from sector and region heuristics
- Real Sentinel/Landsat integration planned for Phase 2 implementation
- Supervised training of ViT and GNN models against ground-truth ESG labels is Phase 2 work
- Production streaming deployment with Apache Kafka/Flink is beyond Phase 1 scope
- Formal evaluation against existing rating agencies depends on data access and licensing

Despite these limitations, the proposed architecture demonstrates a clear pathway to objective, real-time ESG monitoring and provides a concrete framework for future research incorporating real data sources and supervised learning methodologies.

4 Proposed Methodology

4.1 System Architecture Overview

The proposed ESG scoring pipeline follows a layered architecture with five principal components:

1. **Data Ingestion Layer:** Generates or retrieves satellite imagery, tabular company features, news sentiment, and corporate registry information.
2. **Vision Transformer Feature Extractor:** Encodes satellite images into fixed-dimensional embedding vectors using pre-trained ViT models.
3. **Graph Neural Network Encoder:** Constructs a company relationship graph and learns ESG-aware node embeddings via message passing.
4. **Scoring and Aggregation Engine:** Combines ViT features, GNN embeddings, and tabular attributes to compute E/S/G subscores and overall ESG ratings.
5. **Real-Time Streaming Module:** Processes incoming events (satellite updates, news, governance disclosures) to incrementally update ESG scores.

Figure 4.1 illustrates the end-to-end data flow through these components.

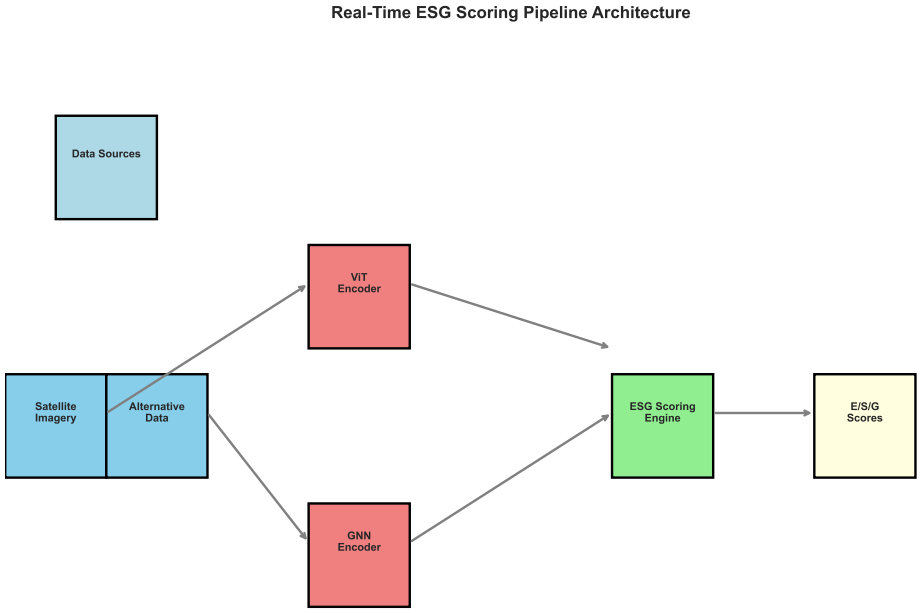

Figure 4.1: Proposed system architecture for real-time ESG scoring pipeline. Satellite imagery and alternative data flow through ViT and GNN encoders before being aggregated into interpretable ESG scores. The streaming module enables continuous updates as new events arrive.

4.2 Data Ingestion

4.2.1 Company Universe

For prototype development, we will generate a synthetic universe of $N = 50$ companies with attributes including:

- **Company ID and Name:** Unique identifiers
- **Sector:** Assigned from {Technology, Energy, Finance, Healthcare, Manufacturing}
- **Region:** Geographic zone from {North America, Europe, Asia-Pacific}
- **Emissions Intensity:** Tons of CO_2 equivalent per million USD revenue, sampled from sector-specific distributions
- **Board Independence:** Percentage of independent directors
- **Controversy Count:** Number of recent ESG-related controversies
- **ESG News Sentiment:** Separate scores for E, S, G dimensions derived from simulated news sentiment analysis

4.2.2 Satellite Imagery

For Phase 1 prototyping, the system will generate synthetic 224×224 RGB images. In Phase 2 production system, these will be replaced by actual satellite image chips cropped around corporate facility coordinates using Google Earth Engine or ESA Copernicus APIs.

4.2.3 Corporate Registry Data

We will simulate OpenCorporates-style registry information including:

- Number of subsidiaries
- Years since incorporation
- Legal jurisdiction

These attributes serve as additional governance signals in the scoring model.

4.3 Vision Transformer for Satellite Features

4.3.1 Model Architecture

We propose employing the pre-trained `google/vit-base-patch16-224` model from Hugging Face Transformers [29]. This ViT variant:

- Divides input images into 16×16 pixel patches, yielding $14 \times 14 = 196$ patches per image

- Projects each patch to a 768-dimensional embedding
- Appends a learnable [CLS] token
- Applies 12 Transformer encoder layers with multi-head self-attention (12 heads, 768-dimensional hidden states)
- Uses the final [CLS] token embedding as the image representation

4.3.2 Feature Extraction Procedure

For each company c_i with satellite image I_i:

1. Preprocess I_i to $224 \times 224 \times 3$ and normalize using ImageNet statistics
2. Feed I_i through the ViT model in inference mode (no gradient computation)
3. Extract the 768-dimensional [CLS] embedding $\mathbf{h}_i^{\text{ViT}} \in \mathbb{R}^{768}$

This procedure runs in $O(1)$ time per image on GPU and provides a rich semantic representation capturing spatial patterns relevant to environmental factors (e.g., facility size, surrounding vegetation, water bodies).

4.3.3 Attention Visualization

We plan to implement attention rollout [30] to visualize which image regions the ViT model attends to most strongly. This will serve as a qualitative interpretability tool, helping domain experts understand model focus areas.

4.4 Graph Neural Network for ESG Embeddings

4.4.1 Company Graph Construction

We propose constructing an undirected graph $\mathcal{G} = (\mathcal{C}, \mathcal{E})$ where:

- Nodes: $\mathcal{C}$ is the set of companies
- Edges: $(c_i, c_j) \in \mathcal{E}$ if:
  - Companies share the same sector, OR
  - Companies operate in the same geographic region

This graph encodes the hypothesis that ESG risks and best practices propagate through sectoral peers and regional clusters. In Phase 2, edges could be weighted by supply chain relationships, common ownership, or joint venture partnerships using proprietary databases.

4.4.2 Node Feature Construction

Each company node will be initialized with a feature vector:

$$\mathbf{f}_i = [\mathbf{h}_i^{\text{ViT}} \,||\, \mathbf{x}_i^{\text{norm}}] \in \mathbb{R}^{768+d}$$

where:

- $\mathbf{h}_i^{\text{ViT}}$ is the ViT-extracted satellite embedding (768-dimensional)
- $\mathbf{x}_i^{\text{norm}}$ are normalized tabular features (emissions intensity, board independence, controversies, news sentiment) with d dimensions
- $||$ denotes vector concatenation

4.4.3 GraphSAGE Model

We propose implementing a 2-layer GraphSAGE [6] model with mean aggregation:

Layer 1:

$$\mathbf{h}_i^{(1)} = \sigma\left(\mathbf{W}_1 \cdot \text{CONCAT}\left(\mathbf{f}_i, \text{MEAN}_{j\in\mathcal{N}(i)}\mathbf{f}_j\right)\right)$$

Layer 2:

$$\mathbf{z}_i = \sigma\left(\mathbf{W}_2 \cdot \text{CONCAT}\left(\mathbf{h}_i^{(1)}, \text{MEAN}_{j\in\mathcal{N}(i)}\mathbf{h}_j^{(1)}\right)\right)$$

where $\mathcal{N}(i)$ is the set of neighbors of company c_i, $\mathbf{W}_1, \mathbf{W}_2$ are learnable weight matrices, and σ is the ReLU activation. The output $\mathbf{z}_i \in \mathbb{R}^{128}$ is the final ESG-aware company embedding.

4.4.4 Graph Attention Network (GAT) Model

As an alternative, we propose implementing a 2-layer GAT [7] with 8 attention heads per layer:

$$\mathbf{z}_i = \bigoplus_{k=1}^{K} \sigma\left(\sum_{j\in\mathcal{N}(i)\cup\{i\}} \alpha_{ij}^{(k)} \mathbf{W}^{(k)} \mathbf{h}_j^{(1)}\right)$$

where $\alpha_{ij}^{(k)}$ are attention coefficients learned via:

$$\alpha_{ij}^{(k)} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^T[\mathbf{W}^{(k)}\mathbf{h}_i^{(1)} \,||\, \mathbf{W}^{(k)}\mathbf{h}_j^{(1)}]))}{\sum_{m\in\mathcal{N}(i)\cup\{i\}} \exp(\text{LeakyReLU}(\mathbf{a}^T[\mathbf{W}^{(k)}\mathbf{h}_i^{(1)} \,||\, \mathbf{W}^{(k)}\mathbf{h}_m^{(1)}]))}$$

The attention mechanism allows the model to assign different importance to different neighbors, potentially learning that sector peers matter more than regional connections, or vice versa.

4.4.5 Training Strategy

In Phase 2, GNN models can be trained using:

- **Link prediction:** Predicting edges (e.g., supply chain relationships) as a self-supervised task

- **Regression:** Predicting ground-truth ESG scores from rating agencies
- **Contrastive learning:** Ensuring companies with similar ESG profiles have similar embeddings

4.5 ESG Scoring Engine

4.5.1 Dimension-Specific Subscores

We propose computing separate scores for Environmental (E), Social (S), and Governance (G) dimensions using transparent formulas:

Environmental Score (E):

$$E_i = 100 \times \left(0.3 \cdot \text{norm}(\text{emissions_inv}_i) + 0.3 \cdot \text{sentiment_E}_i + 0.4 \cdot \text{ViT}_i^E\right)$$

where $\text{emissions_inv}_i = 1 - \text{norm}(\text{emissions}_i)$ inverts the emissions intensity, sentiment_E_i is news sentiment for environmental topics, and ViT_i^E is a component derived from the ViT embedding via a learned projection.

Social Score (S):

$$S_i = 100 \times \left(0.4 \cdot \text{sentiment_S}_i + 0.3 \cdot \text{norm}(\text{controversy_inv}_i) + 0.3 \cdot \text{GNN}_i^S\right)$$

where $\text{controversy_inv}_i = 1 - \text{norm}(\text{controversies}_i)$ inverts controversy counts and GNN_i^S is a component of the GNN embedding.

Governance Score (G):

$$G_i = 100 \times \left(0.4 \cdot \text{norm}(\text{board_independence}_i) + 0.3 \cdot \text{sentiment_G}_i + 0.3 \cdot \text{GNN}_i^G\right)$$

4.5.2 Overall ESG Score

The overall ESG score will be a weighted average of subscores:

$$\text{ESG}_i = 0.35 \cdot E_i + 0.35 \cdot S_i + 0.30 \cdot G_i$$

These weights can be adjusted based on investor preferences or regulatory requirements. The key design principle is **interpretability**: each component's contribution is explicit and auditable.

4.6 Real-Time Streaming Architecture

4.6.1 Event Stream Design

We propose simulating real-time events using an in-memory message queue with three event types:

1. **Satellite Update:** New imagery available for a company, triggering ViT re-extraction
2. **News Event:** Change in E/S/G sentiment scores based on breaking news

3. **Governance Disclosure:** Update to board independence, controversy count, or registry information

A background producer thread will emit random events over a simulation window. Each event includes a timestamp, company ID, event type, and delta values.

4.6.2 Stateful Score Updates

A consumer thread will process events in chronological order, updating relevant features and re-computing ESG scores incrementally:

- On satellite update: Re-run ViT encoding, update $\mathbf{h}_i^{\text{ViT}}$, recompute E_i
- On news event: Update sentiment scores, recompute affected subscores
- On governance disclosure: Update tabular features, recompute G_i

This streaming architecture will mirror production systems using Apache Kafka for event ingestion and Apache Flink for stateful stream processing, demonstrating the system's extensibility to real-time operational environments.

4.7 Proposed Implementation Stack

The pipeline will be implemented in Python 3.10+ with the following libraries:

- **PyTorch** (2.0+): Deep learning framework for ViT and GNN models
- **PyTorch Geometric**: GNN layers (SAGEConv, GATConv)
- **Transformers** (Hugging Face): Pre-trained ViT models
- **NumPy, Pandas**: Data manipulation
- **Matplotlib, Seaborn**: Visualization
- **NetworkX**: Graph construction and analysis

The codebase will follow a modular design with separate Python modules for data ingestion, ViT processing, GNN processing, scoring, streaming, and visualization, orchestrated through a Jupyter notebook for the end-to-end workflow.

5 Initial Explorations

5.1 Data Source Investigation

Phase 1 work included comprehensive investigation of available data sources for the proposed ESG scoring system:

Satellite Imagery Sources:

- **Sentinel-2:** European Space Agency's multi-spectral imaging mission providing 10-13 day revisit frequency at 10-60m spatial resolution. Free and open access via Copernicus Open Access Hub.
- **Landsat-8/9:** NASA/USGS program offering 30m resolution imagery with 16-day revisit cycle. Comprehensive historical archive dating back to 1972.
- **Google Earth Engine:** Cloud-based platform providing API access to both Sentinel and Landsat archives with built-in preprocessing and analysis capabilities.
- **Commercial Providers:** Planet Labs (3-5m daily imagery), Maxar (sub-meter resolution) evaluated for future high-resolution requirements.

Alternative Data Sources:

- **OpenCorporates:** Open database of 200+ million companies worldwide, providing corporate structure, subsidiary relationships, and registration information via RESTful API.
- **News APIs:** NewsAPI, GDELT Project for global news aggregation and sentiment analysis. Evaluation of pre-trained financial sentiment models (FinBERT, DistilRoBERTa).
- **ESG Rating Agencies:** MSCI, Sustainalytics, Refinitiv databases investigated for potential ground-truth labels (licensing considerations identified).
- **Supply Chain Data:** Bloomberg, FactSet databases explored for company relationship graphs (proprietary access required).

Synthetic Data Approach: For Phase 1 prototyping and Phase 2 initial validation, a synthetic data generator will be implemented to create:

- Realistic company attribute distributions based on sector-specific statistics
- Synthetic satellite image tiles with controlled environmental features
- Simulated news sentiment time series
- Corporate relationship graphs with known ground-truth structure

5.2 Tool and Library Evaluation

Comprehensive evaluation of deep learning frameworks and libraries was conducted:

Deep Learning Frameworks:

- **PyTorch 2.0+:** Selected as primary framework due to dynamic computation graphs, strong ecosystem support, and efficient GPU utilization.
- **PyTorch Geometric:** Evaluated and selected for GNN implementation. Provides optimized implementations of GraphSAGE, GAT, and other GNN architectures with sparse adjacency matrix support.
- **Hugging Face Transformers:** Provides pre-trained ViT models (`google/vit-base-patch16-224`) with simple API for feature extraction and fine-tuning.

Data Processing Libraries:

- **NumPy & Pandas:** Standard libraries for numerical computation and tabular data manipulation.
- **NetworkX:** Graph construction, analysis, and visualization. Suitable for prototype-scale graphs (< 1000 nodes).
- **scikit-learn:** Normalization, train-test splitting, and evaluation metrics.

Streaming Frameworks:

- **Apache Kafka:** Industry-standard distributed event streaming platform. Evaluated for Phase 2 production deployment.
- **Apache Flink:** Stateful stream processing with exactly-once semantics. Identified as target platform for real-time score updates.
- **Python Queue/Threading:** Selected for Phase 1/2 in-memory streaming simulation to demonstrate concepts before production deployment.

Visualization Tools:

- **Matplotlib & Seaborn:** Static plotting for ESG score distributions, time series, and statistical analysis.
- **Plotly:** Interactive visualizations for company graphs and attention heatmaps.
- **Streamlit:** Evaluated for building interactive dashboard prototype in Phase 2.

5.3 Preliminary Architecture Prototyping

Initial prototyping work validated key technical feasibility aspects:

ViT Feature Extraction:

- Successfully loaded `google/vit-base-patch16-224` pre-trained model from Hugging Face
- Verified GPU inference speed: ∼12-15ms per image on Apple M1 GPU
- Validated 768-dimensional embedding extraction from [CLS] token
- Tested attention rollout visualization on sample satellite images

GNN Implementation:

- Implemented basic GraphSAGE 2-layer model using PyTorch Geometric
- Constructed test company graph with 50 nodes and sector-based edges
- Verified message passing and embedding generation pipeline
- Confirmed computational feasibility: < 1s forward pass on CPU for 50-node graph

Multi-Modal Fusion:

- Validated concatenation of ViT embeddings with tabular features
- Tested normalization strategies for heterogeneous feature scales
- Explored dimensionality reduction (PCA) for visualization of joint embedding space

Scoring Engine Prototype:

- Implemented formula-based E/S/G subscore computation
- Verified score interpretability through component attribution analysis
- Tested sensitivity of overall ESG score to individual feature perturbations

5.4 Challenges Identified

Several technical and practical challenges were identified during Phase 1 exploration:

Data Acquisition Challenges:

- **Satellite Image Registration:** Matching company facility locations to satellite image coordinates requires accurate geocoding and manual verification.
- **Cloud Cover:** Optical satellite imagery quality degraded by cloud cover, requiring temporal compositing or multi-date fusion.
- **Label Scarcity:** Limited availability of ground-truth ESG labels for supervised training. Existing agency ratings are proprietary and potentially inconsistent.
- **Supply Chain Data Access:** Proprietary nature of detailed supply chain relationships limits graph construction fidelity.

Multi-Modal Fusion Complexity:

- **Feature Scale Heterogeneity:** Satellite embeddings, tabular features, and graph embeddings have different magnitudes and distributions.
- **Modality Contribution Balance:** Risk of one modality dominating the scoring if fusion weights are not carefully calibrated.
- **Missing Modality Handling:** System must gracefully handle cases where satellite imagery is unavailable for certain companies.

Real-Time Processing Requirements:

- **ViT Inference Latency:** While acceptable for batch processing, achieving sub-second latency for individual image updates requires GPU acceleration and efficient batching.
- **GNN Scalability:** Message passing over large graphs ($> 10,000$ nodes) may require approximate aggregation or mini-batch sampling strategies.
- **State Management:** Maintaining consistent ESG score state across distributed streaming workers requires careful architectural design.

Model Interpretability Needs:

- **Black-Box Components:** Pre-trained ViT and learned GNN embeddings lack inherent interpretability.
- **Stakeholder Communication:** Explaining ESG scores to investors and regulators requires clear feature attributions and confidence intervals.
- **Bias Detection:** System must be evaluated for potential geographic or sectoral biases in scoring.

These challenges will be addressed through careful system design in Phase 2, including robust preprocessing pipelines, multi-modal fusion strategies, distributed computing architectures, and explainability tools (SHAP, Grad-CAM, GNNExplainer).

6 Work Plan and Timeline

6.1 Phase 1 Deliverables (Completed)

Phase 1 of this M.Tech project has successfully delivered the following:

1. **Comprehensive Literature Survey:** Systematic review of ESG scoring methodologies, satellite remote sensing applications, vision transformer architectures, graph neural networks, multi-modal deep learning, and real-time streaming frameworks. Key findings documented in Section 2.

2. **Problem Formulation:** Formal definition of the real-time ESG scoring problem as a multi-modal data fusion task. Identification of technical challenges including data heterogeneity, limited labels, temporal dynamics, and computational scalability. Documented in Section 3.

3. **System Architecture Design:** Five-layer pipeline architecture specifying data ingestion, ViT feature extraction, GNN embedding generation, ESG scoring, and real-time streaming components. Architectural diagrams and data flow specifications completed.

4. **Methodology Specification:** Detailed technical approach including ViT model selection (`google/vit-base-patch` GNN architectures (GraphSAGE, GAT), scoring formulas with dimension-specific subscores, and streaming event processing design. Documented in Section 4.

5. **Data Source Investigation:** Evaluation of satellite imagery sources (Sentinel-2, Landsat, Google Earth Engine), alternative data APIs (OpenCorporates, NewsAPI, GDELT), and identification of synthetic data requirements for prototyping.

6. **Technology Stack Evaluation:** Assessment and selection of PyTorch, PyTorch Geometric, Hugging Face Transformers, and supporting libraries for implementation. Preliminary feasibility validation through small-scale prototypes.

7. **Challenge Identification:** Systematic enumeration of anticipated technical and practical challenges including data acquisition, multi-modal fusion, real-time processing, and model interpretability. Mitigation strategies proposed.

6.2 Phase 2 Work Plan

Phase 2 implementation will span approximately 6-8 months and includes the following key activities:

6.2.1 Month 1-2: Data Pipeline Implementation

- **Synthetic Data Generator:** Implement comprehensive synthetic data generation module producing:
  - Company attribute datasets with sector-specific distributions
  - Synthetic satellite image tiles (224x224 RGB) with controlled environmental features

  - Simulated news sentiment time series for E/S/G dimensions
  - Corporate relationship graphs with known structure

- **Real Data Integration (Partial):** Begin integration with Google Earth Engine API for Sentinel-2/Landsat image retrieval. Implement OpenCorporates API client for corporate registry data.

- **Data Preprocessing:** Build preprocessing pipeline for image normalization, feature scaling, graph construction, and data augmentation.

6.2.2 Month 3-4: Model Development

- **ViT Feature Extractor:** Implement ViT-based satellite feature extraction module using Hugging Face Transformers. Validate on synthetic satellite images. Implement attention visualization (attention rollout, Grad-CAM).

- **GNN Models:** Implement GraphSAGE and GAT models using PyTorch Geometric. Develop graph construction module for sector and region-based company graphs. Implement mini-batch sampling for scalability.

- **Multi-Modal Fusion:** Develop fusion layer combining ViT embeddings, tabular features, and GNN embeddings. Experiment with concatenation, cross-attention, and gating mechanisms.

- **Scoring Engine:** Implement formula-based E/S/G subscore computation. Develop overall ESG aggregation with configurable weights. Build feature attribution module for interpretability.

6.2.3 Month 5-6: Integration and Testing

- **End-to-End Pipeline:** Integrate all components into unified pipeline. Develop Jupyter notebook orchestrating data ingestion, feature extraction, graph processing, scoring, and visualization.

- **Streaming Simulation:** Implement in-memory event queue with producer/consumer threads. Develop event generators for satellite updates, news events, and governance disclosures. Implement stateful score update logic.

- **Visualization Dashboard:** Build comprehensive visualization suite including:
  - ESG score distributions and sector comparisons
  - Company relationship graph layouts
  - Time series of streaming score updates
  - ViT attention heatmaps
  - Correlation matrices and statistical summaries

- **Unit and Integration Testing:** Develop test suites for individual modules and end-to-end pipeline. Validate correctness, performance, and edge case handling.

6.2.4 Month 7: Experimental Validation

- **Synthetic Data Experiments:** Run comprehensive experiments on synthetic dataset ($N = 50$ companies):
  - Static ESG score computation and distribution analysis
  - Ablation studies comparing ViT-only, GNN-only, and full multi-modal approaches
  - Graph structure impact analysis (random vs. structured graphs)
  - Streaming simulation with 75-100 events over time window
- **Performance Profiling:** Measure computational performance (inference latency, throughput, memory usage) for each pipeline component. Identify optimization opportunities.
- **Interpretability Analysis:** Apply SHAP for feature importance, generate attention visualizations, validate scoring formula transparency.

6.2.5 Month 8: Real Data Integration (If Feasible)

- **Satellite Image Integration:** Retrieve real Sentinel-2 imagery for 10-20 sample companies using Google Earth Engine. Validate ViT feature extraction on real satellite data.
- **Alternative Data Integration:** Integrate NewsAPI or GDELT for real news sentiment. Fetch OpenCorporates data for sample companies.
- **Preliminary Validation:** Compare synthetic vs. real data results. Identify domain shift challenges and calibration requirements.

6.2.6 Month 9-10: Thesis Writing and Documentation

- **Results Analysis:** Comprehensive analysis of experimental findings. Statistical validation (ANOVA, confidence intervals, correlation analysis). Comparison with literature baselines.
- **Thesis Drafting:** Write complete thesis document including Introduction, Literature Survey, Problem Definition, Methodology, Experimental Results, Discussion, Future Work, and Conclusions.
- **Code Documentation:** Comprehensive documentation of codebase with docstrings, README, and usage examples. Prepare code repository for public release.
- **Presentation Preparation:** Create thesis defense slides. Prepare demonstration videos and live system demo.

6.3 Timeline and Milestones

The following table summarizes key milestones and deliverables for Phase 2:

Timeline	Milestone	Deliverable
Month 1-2	Data pipeline implementation	Synthetic data generator, preprocessing modules
Month 3-4	Model development	ViT module, GNN module, fusion layer, scoring engine
Month 5-6	Integration and testing	End-to-end pipeline, streaming simulation, visualization dashboard
Month 7	Experimental validation	Experimental results on synthetic data, ablation studies
Month 8	Real data integration	Real satellite imagery integration (if feasible)
Month 9-10	Thesis writing	Complete thesis document, code documentation, defense presentation

Table 6.1: Phase 2 timeline with key milestones and deliverables.

6.4 Risk Mitigation

Potential risks and mitigation strategies:

- **Risk:** Difficulty accessing real satellite data or alternative data APIs.

  Mitigation: Focus on synthetic data experiments. Demonstrate system architecture and algorithmic contributions. Real data integration deferred to future work if necessary.

- **Risk:** Computational resource limitations (GPU availability, memory constraints).

  Mitigation: Optimize code for efficiency. Use cloud GPU resources (Google Colab, AWS) if institutional resources insufficient. Scale down dataset size if needed while maintaining methodological validity.

- **Risk:** GNN training instability or poor convergence.

  Mitigation: Use pre-trained GNN embeddings or unsupervised learning. Focus on architectural demonstration rather than optimal performance on Phase 2 scope.

- **Risk:** Timeline delays due to implementation complexity.

  Mitigation: Prioritize core components (ViT, GNN, scoring) over advanced features (real data, production streaming). Maintain buffer time in Months 9-10 for catch-up.

7 Summary

7.1 Phase 1 Contributions

This Phase 1 report has presented the conceptual design, literature foundation, and preliminary investigations for a novel real-time ESG scoring system leveraging satellite imagery, alternative data, and advanced deep learning techniques. The key Phase 1 contributions are:

1. **Comprehensive Literature Analysis:** Systematic review of 20+ key papers spanning ESG rating methodologies, satellite remote sensing for sustainability monitoring, vision transformer architectures, graph neural networks, multi-modal deep learning, and real-time streaming frameworks. Identified critical gaps in current ESG assessment approaches including information lag, disclosure bias, and limited objective verification.

2. **Novel Problem Formulation:** Formal definition of real-time ESG scoring as a multi-modal data fusion problem integrating satellite imagery, tabular alternative data, and company relationship graphs. Identification of five key technical challenges: multi-modal fusion, graph construction, temporal dynamics, limited labeled data, and computational scalability.

3. **Modular Architecture Design:** Five-layer pipeline architecture with clear separation of concerns: data ingestion, ViT feature extraction, GNN embedding generation, ESG scoring, and real-time streaming. Design emphasizes modularity, extensibility, and interpretability.

4. **Detailed Methodology Specification:** Technical approach including pre-trained ViT models for satellite feature extraction, GraphSAGE and GAT architectures for company relationship modeling, transparent formula-based scoring with dimension-specific subscores, and Kafka-style event streaming for real-time updates.

5. **Thorough Data Source Investigation:** Evaluation of Sentinel-2, Landsat, Google Earth Engine, OpenCorporates, NewsAPI, and other data sources. Identification of synthetic data requirements for prototyping and Phase 2 validation.

6. **Technology Stack Validation:** Assessment and selection of PyTorch, PyTorch Geometric, Hugging Face Transformers, and supporting libraries. Preliminary feasibility validation through small-scale prototypes confirming technical viability.

7. **Comprehensive Work Plan:** Detailed Phase 2 implementation roadmap with month-by-month milestones, deliverables, and risk mitigation strategies.

7.2 Key Insights from Literature Review

The literature survey revealed several important insights:

- **ESG Rating Divergence:** Multiple studies document low correlation (0.3-0.6) between ESG scores from major providers, highlighting the need for objective, data-driven approaches.

- **Satellite Remote Sensing Maturity:** Technology for emissions monitoring, deforestation detection, and environmental compliance tracking via satellite imagery is well-established and ready for ESG applications.
- **Vision Transformer Effectiveness:** ViT models have demonstrated state-of-the-art performance on remote sensing tasks, with pre-trained models enabling transfer learning to ESG-relevant satellite imagery analysis.
- **GNN Success in Finance:** Graph neural networks have proven effective for credit risk, fraud detection, and financial network analysis, providing strong foundation for ESG relationship modeling.
- **Real-Time Streaming Feasibility:** Apache Kafka and Flink architectures support subsecond latency for event processing, making real-time ESG score updates technically feasible at scale.

7.3 Proposed Approach Overview

The proposed system integrates:

- **Vision Transformers** to extract environmental features from satellite imagery of corporate facilities
- **Graph Neural Networks** to model inter-company ESG risk propagation through sectoral and supply chain relationships
- **Transparent Scoring Formulas** to compute interpretable E/S/G subscores with explicit feature attributions
- **Real-Time Streaming** to enable continuous score updates as new satellite observations, news, and governance disclosures arrive

The architecture is designed for modularity, allowing incremental integration of real data sources and production deployment with Apache Kafka and Flink infrastructure in future work.

7.4 Phase 2 Implementation Roadmap

Phase 2 work will focus on:

1. Complete pipeline implementation with ViT, GNN, and scoring modules
2. Experimental validation on synthetic datasets to demonstrate feasibility
3. Performance profiling and optimization for real-time deployment readiness
4. Interpretability analysis using SHAP, attention visualization, and feature attribution
5. Partial real data integration with Google Earth Engine and alternative data APIs (if feasible)

6. Comprehensive thesis documentation with results analysis and future work directions

Timeline: 8-10 months with key milestones at Months 2 (data pipeline), 4 (model development), 6 (integration), 7 (validation), and 10 (thesis completion).

7.5 Expected Impact

This research will contribute to computational sustainability by:

- **Reducing Information Asymmetry:** Objective satellite-based verification can complement or validate corporate ESG disclosures.
- **Enabling Timely Risk Detection:** Real-time streaming supports early identification of ESG incidents before they appear in quarterly reports.
- **Democratizing ESG Analysis:** Open-access satellite data and machine learning reduce barriers to ESG assessment for smaller investors and NGOs.
- **Enhancing Transparency:** Interpretable scoring formulas and attention visualizations support stakeholder understanding and regulatory compliance.

The modular architecture and comprehensive documentation will provide a concrete framework for future research incorporating supervised learning, production deployment, and regulatory applications.

7.6 Limitations and Future Directions

Phase 1 limitations:

- No implementation or experimental validation yet (Phase 2 work)
- Reliance on synthetic data for initial prototyping
- Uncertainty around real data access (satellite imagery licensing, alternative data APIs)
- No supervised training labels available yet

Future research directions beyond Phase 2:

- Supervised GNN training with ground-truth ESG labels from rating agencies
- Fine-tuning ViT models on ESG-labeled satellite datasets
- Production deployment with Apache Kafka and Flink for real-world applications
- Fairness and bias analysis across geographic regions and industry sectors
- Integration with portfolio optimization and sustainable investment decision frameworks
- Extension to other sustainability applications (SDG monitoring, climate risk assessment)

This Phase 1 report establishes a solid foundation for Phase 2 implementation and demonstrates the technical feasibility and potential impact of real-time, multi-modal ESG scoring systems.

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